Smart Street Light – Arduino Code

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// Smart Street Light
// Wiring:
//   LED 1 -> D5 (through 220 ohm)
//   LED 2 -> D6 (through 220 ohm)
//   LDR divider: 5V -> LDR -> A0 -> 10K -> GND  (brighter light = higher A0 value)
//   IR sensor 1 OUT -> D2
//   IR sensor 2 OUT -> D3

const int LED1_PIN = 5;
const int LED2_PIN = 6;
const int IR1_PIN  = 2;
const int IR2_PIN  = 3;
const int LDR_PIN  = A0;

// ---- Settings you can tune ----
const int DARK_BELOW   = 400;   // LDR value below this = dark (lights turn on)
const int BRIGHT_ABOVE = 500;   // LDR value above this = daylight (lights turn off)
const int DIM_LEVEL    = 40;    // 0-255, brightness when dark and no motion
const int FULL_LEVEL   = 255;   // brightness when motion is detected
const unsigned long HOLD_MS = 3000;  // stay bright this long after last detection

// Most IR obstacle modules output LOW when they detect something.
// If yours outputs HIGH on detection, change this to HIGH.
const int DETECTED_STATE = LOW;

bool isDark = false;
unsigned long lastDetect1 = 0;
unsigned long lastDetect2 = 0;
bool detected1Ever = false;
bool detected2Ever = false;

void setup() {
  pinMode(LED1_PIN, OUTPUT);
  pinMode(LED2_PIN, OUTPUT);
  pinMode(IR1_PIN, INPUT);
  pinMode(IR2_PIN, INPUT);
  Serial.begin(9600);
}

void loop() {
  int ldr = analogRead(LDR_PIN);

  // Hysteresis so the lights don't flicker around the threshold
  if (ldr < DARK_BELOW)        isDark = true;
  else if (ldr > BRIGHT_ABOVE) isDark = false;

  unsigned long now = millis();

  if (digitalRead(IR1_PIN) == DETECTED_STATE) {
    lastDetect1 = now;
    detected1Ever = true;
  }
  if (digitalRead(IR2_PIN) == DETECTED_STATE) {
    lastDetect2 = now;
    detected2Ever = true;
  }

  bool active1 = detected1Ever && (now - lastDetect1 < HOLD_MS);
  bool active2 = detected2Ever && (now - lastDetect2 < HOLD_MS);

  if (isDark) {
    analogWrite(LED1_PIN, active1 ? FULL_LEVEL : DIM_LEVEL);
    analogWrite(LED2_PIN, active2 ? FULL_LEVEL : DIM_LEVEL);
  } else {
    analogWrite(LED1_PIN, 0);
    analogWrite(LED2_PIN, 0);
  }

  Serial.print("LDR: ");
  Serial.print(ldr);
  Serial.print("  Dark: ");
  Serial.print(isDark ? "YES" : "NO");
  Serial.print("  IR1: ");
  Serial.print(active1 ? "ON" : "off");
  Serial.print("  IR2: ");
```

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  Serial.println(active2 ? "ON" : "off");

  delay(50);
}
```